

YAN DAI

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✉ yandai20@mit.edu

EDUCATION

Massachusetts Institute of Technology Operations Research Center 2024 — Now

PhD student in Operations Research co-advised by Patrick Jaillet and Negin Golrezaei. GPA: 5.0 / 5.

Tsinghua University Andrew C. Yao Class 2020 — 2024

Bachelor of Engineering in Computer Science and Technology. GPA: 3.99/4.00.

RESEARCH INTERESTS

EconCS: market design, pricing, and resource allocation with strategic agents and incomplete information.

Learning Theory: bandits, online learning, and adversarial reinforcement learning theory.

PUBLICATIONS AND WORKING PAPERS

Market Design, Pricing, and Resource Allocation

[15] **[NeurIPS'25]** [Yan Dai](#), Negin Golrezaei, and Patrick Jaillet. “Efficiency, Feasibility, and Incentive-Awareness in Constrained Online Resource Allocation.” **Under Review at Operations Research**.

1st Place in ACM Student Research Competition (SRC), SIGMETRICS'25.

[14] **[COLT'25]** [Yan Dai](#), Moïse Blanchard, and Patrick Jaillet. “Non-Monetary Mechanism Design without Priors: Achieving Efficiency via Adaptive Costly Audits.” **Major Revision at Operations Research**.

[13] [Yan Dai](#), Maryam Farboodi, Negin Golrezaei, and Sepehr Shahshahani. “Market Design for AI: Beyond the Copyright Binary.” Early version accepted to 2026 M&SOM Service Operations SIG (acceptance: 10/96).

Bandits and Online Learning Theory

[12] **[SIGMETRICS'25]** [Yan Dai](#) and Longbo Huang. “Adversarial Network Optimization under Bandit Feedback: Maximizing Utility in Non-Stationary Multi-Hop Networks.”

Best Paper Award of ACM SIGMETRICS'25.

[11] **[ICLR'25]** Yu Chen*, Jiatai Huang*, [Yan Dai*](#), and Longbo Huang. “uniINF: Best-of-Both-Worlds Algorithm for Parameter-Free Heavy-Tailed MABs.” *Spotlight (top 5%)*.

[10] **[ICML'23]** Jiatai Huang*, [Yan Dai*](#), and Longbo Huang. “Banker Online Mirror Descent: A Universal Approach for Delayed Online Bandit Learning.”

[9] **[ICLR'23]** [Yan Dai](#), Ruosong Wang, and Simon S. Du. “Variance-Aware Sparse Linear Bandits.”

[8] **[ICML'22]** Jiatai Huang*, [Yan Dai*](#), and Longbo Huang. “Adaptive Best-of-Both-Worlds Algorithm for Heavy-Tailed Multi-Armed Bandits.”

[7] [Yan Dai](#), Negin Golrezaei, and Patrick Jaillet. “Policy Regret for Embedding Model Routing: Contextual Bandits with Low-Rank Experts.”

Adversarial Reinforcement Learning Theory

[6] **[COLT'24]** [Yan Dai](#), Qiwen Cui, and Simon S. Du. “Refined Sample Complexity for Markov Games with Independent Linear Function Approximation.”

[5] **[ICML'23]** [Yan Dai](#), Haipeng Luo, Chen-Yu Wei, and Julian Zimmert. “Refined Regret for Adversarial MDPs with Linear Function Approximation.”

[4] **[NeurIPS'22]** [Yan Dai](#), Haipeng Luo, and Liyu Chen. “Follow-the-Perturbed-Leader for Adversarial Markov Decision Processes with Bandit Feedback.”

[3] Aarush Kulkarni, Khang Nguyen, Ricardo Parada, Kenny Guo, William Chang, and [Yan Dai](#). “Learning Adversarial Continuous MDPs with Bandit Feedback and Unknown Transitions.” *Mentoring Project*.

Additional Work on Non-Convex Optimization Theory

- [2] **[ICML'24]** Kwangjun Ahn, Zhiyu Zhang, Yunbum Kook, and [Yan Dai](#). “Understanding Adam Optimizer via Online Learning of Updates: Adam is FTRL in Disguise.”
- [1] **[NeurIPS'23]** [Yan Dai*](#), Kwangjun Ahn*, and Suvrit Sra. “The Crucial Role of Normalization in Sharpness-Aware Minimization.”

SELECTED AWARDS

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| • Runner-Up of Two Sigma PhD Fellowship (<i>2 winners and 2 runners-up</i>). | Feb 2026 |
| • Best Paper Award (<i>1 paper per year</i>) of ACM SIGMETRICS 2025. | Jun 2025 |
| • 1st Place in ACM Student Research Competition (SRC) held at SIGMETRICS 2025. | Jun 2025 |
| • Outstanding Graduate named by Beijing (<i>5%</i>), by Tsinghua (<i>2%</i>), and by Yao Class (<i>10%</i>). | Jun 2024 |
| • Presidential Scholarship (<i>top scholarship for Tsinghua undergrads; 10/16,000+</i>), Tsinghua. | Nov 2023 |
| • Andrew C. Yao Award – Gold Medal (<i>top scholarship for seniors; 1 student</i>), Yao Class. | Sep 2023 |
| • Gold Medal (<i>1st place with perfect score</i>), Asia-Pacific Informatics Olympiad (APIO). | May 2019 |
| • Gold Medal (<i>5th place</i>), Chinese National Olympiad in Informatics (NOI). | Jul 2018 |
| • Gold Medal, Asia-Pacific Informatics Olympiad (APIO). | May 2018 |
| • Gold Medal, IOI China Team Selection Competition (CTSC). | May 2018 |

EXPERIENCE

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|---|----------------|
| Massachusetts Institute of Technology OR Center (Cambridge, MA)
Research assistant of Patrick Jaillet and Negin Golrezaei. | Sep 2024 – Now |
| Microsoft Research EconCS Group (Cambridge, MA)
Research intern mentored by Alex Slivkins and Brendan Lucier (also worked closely with Nicole Immorlica). | May – Aug 2026 |
| Tsinghua University Andrew C. Yao Class (Beijing, China)
Bachelor’s thesis advised by Longbo Huang. | Feb – Jun 2024 |
| University of Washington School of CSE (Seattle, WA)
Visiting student hosted by Simon S. Du. | Jun – Aug 2023 |
| Massachusetts Institute of Technology Lab for Info. & Decision (Cambridge, MA)
Visiting student hosted by Suvrit Sra. | Feb – May 2023 |
| University of Southern California CS Department (Remote)
Remote visitor of Haipeng Luo. | Feb – Dec 2022 |
| Tsinghua University Andrew C. Yao Class (Beijing, China)
Research assistant of Longbo Huang. | Jun – Dec 2021 |

PROFESSIONAL SERVICES

Journal Reviewer.

- **Operations Research & Economics:** *Operations Research; Games and Economic Behavior* (GEB).
- **Machine Learning:** *Journal of Machine Learning Research* (JMLR), *IEEE Transactions on Pattern Analysis and Machine Intelligence* (TPAMI), *Machine Learning, Transactions on Machine Learning Research* (TMLR).
- **Networking & Systems:** *IEEE/ACM Transactions on Networking* (TON), *Performance Evaluation* (PEVA), *IEEE Internet of Things Journal* (IoT-J).

Conference Reviewer. COLT 2025; ICML 2026 (gold reviewer) / 2025 / 2023; NeurIPS 2026 / 2025 / 2024 / 2023 / 2022; ICLR 2027 / 2025 / 2024; AISTATS 2027 / 2026 / 2025 / 2023 / 2022; AAAI 2027 / 2026; ALT 2023.